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Milestone 1 — Predicting Ambulance Response Times in Dublin Using Machine Learning

PROJECT AIM

The aim of this project is to develop a machine-learning system that predicts the response time of an ambulance to an incident in Dublin. Response time will be defined as the time between TOC (Time of Call) and IA (Incident Arrived).

The main research question is:

Can temporal, geographic, incident-priority and historical-demand features be used to predict ambulance response times in Dublin?

This will be treated as a supervised regression problem. Candidate models will include a simple baseline, Linear Regression, Random Forest and Gradient Boosting/XGBoost. Performance will primarily be measured using Mean Absolute Error (MAE), with root mean squared error and R^2 used as additional metrics where appropriate.

The project will also investigate whether adding increasingly detailed features improves prediction performance, for example comparing a basic temporal model against models incorporating geographic and historical-demand information.

RAW DATA SOURCE

The raw data will come from the official Dublin Fire Brigade and Ambulance dataset published on Ireland's open-data portal, data.gov.ie: *https://data.gov.ie/dataset/fire-brigade-and-ambulance*

The dataset contains ambulance incident records for the Greater Dublin Area and includes operational timestamps such as:

- TOC: Time of Call
- ORD: vehicle ordered
- MOB: vehicle mobile
- IA: arrived at incident
- LS: left scene
- AH: arrived at hospital
- MAV: vehicle available
- CD: incident closed

It also contains incident classifications, priority information and geographic/operational information that can be used to construct predictive features.

The dataset contains substantially more than the required 10,000 learning samples. Individual annual ambulance datasets contain tens of thousands of records, allowing multiple years to be combined for model development.

RAW DATA STORAGE

The original downloaded files will be stored unchanged in a local project directory:

`data/raw/`

The raw files will not be modified after download. Each source file will retain information about its source year/resource in its filename, for example:

`data/raw/ambulance_2025.csv`

This preserves the original data so that the processed dataset can always be regenerated from the source data.

The project repository will also contain a model card documenting the official data.gov.ie source URL and the date on which each resource was downloaded.

PROCESSED DATA STORAGE AND FILE FORMATS

After preprocessing, cleaned and feature-engineered datasets will be stored separately from the raw data in:

`data/processed/`

The original source data is CSV, while processed tabular datasets will preferably be stored as Parquet files because Parquet preserves column data types and is efficient for repeated analytical/model-training operations.

For example:

`data/processed/ambulance_clean.parquet`

and, where useful:

`data/processed/train.parquet`
`data/processed/validation.parquet`
`data/processed/test.parquet`

The processed files will contain the cleaned target variable and model features rather than modifying the original source files.

DATABASE / OBJECT STORAGE DECISION

A database is not required for the initial project. The dataset is a relatively manageable collection of historical tabular records and the project mainly requires sequential batch processing and model training rather than high-frequency transactional queries.

A local file-based structure using CSV for immutable raw data and Parquet for processed data is therefore appropriate for this project.

If the dataset becomes too large for efficient local processing, object storage or a database could be considered, but this is not expected to be necessary for the project's current scope.

DATA VERSIONING

Data versions will be identified using the source resource/year, download date and, where appropriate, a file checksum.

For example:

`ambulance_2025_YYYYMMDD.csv`

The project README or a metadata file will record:

- Official source URL
- Resource/year
- Download date
- Original filename
- Number of rows
- Number of columns
- Any important changes between source releases

The raw files will be retained so that a particular processed dataset can be traced back to the exact source data used to generate it.

DATA ACCESS

The data will be downloaded from the official data.gov.ie resource and accessed programmatically using Python.

The preprocessing pipeline will read the raw CSV files from:

`data/raw/`

and produce processed datasets in:

`data/processed/`

No credentials are required for the public dataset. The source URL and download information will be documented so another student can obtain the same source data.

The project will use Python tools such as pandas or Polars for data processing and scikit-learn-compatible structures for machine-learning experiments.

Code will be included in the next project milestone.

TARGET VARIABLE

The target variable will be ambulance response time, measured in minutes.

It will be calculated from the incident timestamps as:

Response Time = IA – TOC

The resulting target is a continuous numerical variable, making this a regression problem.

Records with missing or unusable TOC/IA timestamps will need to be investigated and handled.

FEATURES

Potential input features will include:

Incident characteristics
- Incident priority/criticality
- Incident type or classification, where available

Temporal features

- Hour of day
- Day of week
- Month
- Season
- Weekend indicator
- Rush-hour indicator

Geographic features
- Incident location or geographic area
- Station/location information where available
- Potential distance-related features if reliable station

GitHub repository (currently empty): https://github.com/Kate123Rich/ai-infra